BIKE STAND ANALYSIS

1. Project Overview and Objective

Overview:

This project involves cleaning, transforming, and analysing raw data using Excel and creating an interactive Power BI dashboard to derive meaningful business insights.

Object:

The main objective is to demonstrate data pre-processing techniques using an interactive Power BI dashboard visualization to make informed decisions.

2. Data Sources

- **Source Description and Timeline:**
  https://github.com/GeethaGunasekaran1/Project-Dataset/raw/refs/heads/main/bike-stations-sharing-data.xlsx.
- Domain: sales

3. Problem Statement;

- To analyse sales performance across products to identify high performing areas and profit trends.
- To understand customer purchasing behaviour to enhance marketing and retention strategies.

4. Attribute (Column /Features) Details:

Column Name	Description
Contract id	Unique identifier for the contract or network operator
Contract Name	Name of the contract or operator system

Column Name	Description
number	Unique identification number of the bike station
name	Descriptive name of the bike station
address	Street address or location description of the station
City	City where the station is located
banking	Indicates if a credit card/banking terminal is available (`True` / `False`)
bonus	Indicates if the station participates in a bonus reward program (`True` / `False`)
status	Operational status of the station (`OPEN` / `CLOSED`)
Bike Stands	Total capacity / total number of bike docks at the station
Available Bike Stands	Number of empty docks available to return a bike
Available Bikes	Number of bicycles currently available for rent
Last Update	Timestamp of when the station data was last refreshed

5. Tools & Technologies

Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard.

6. Data Pre-Processing (Excel / Power Query)

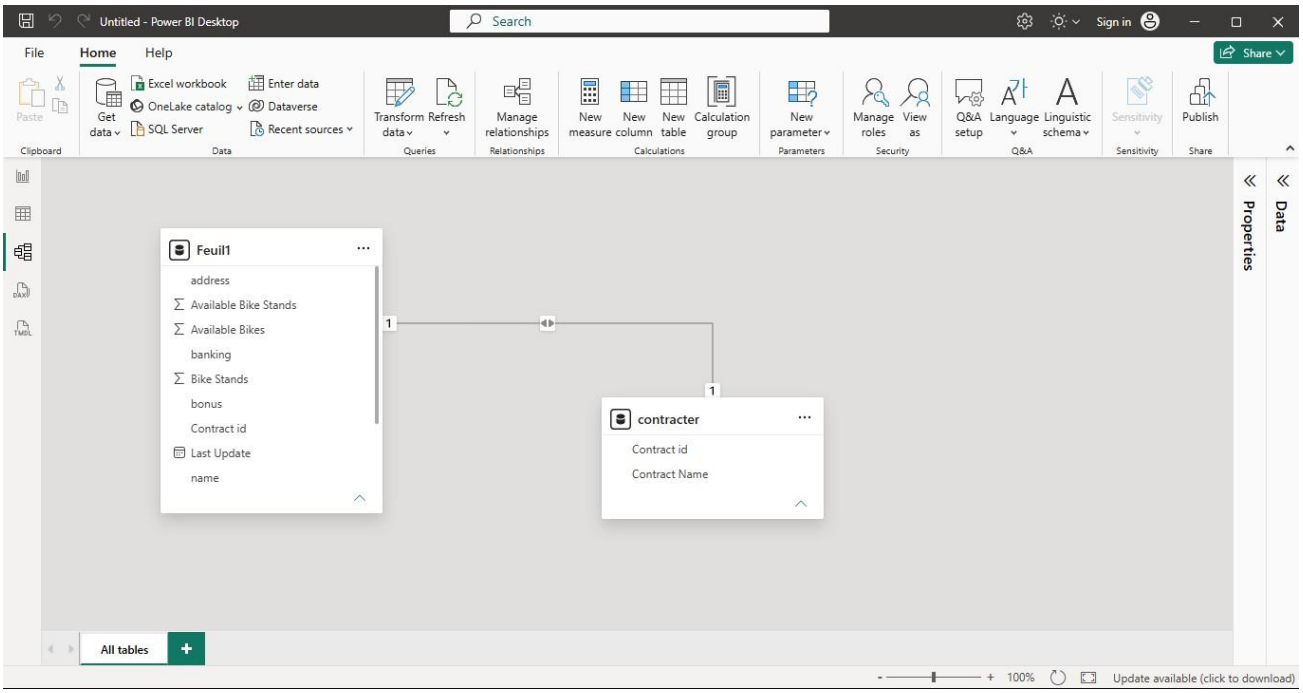
Tasks Performed:

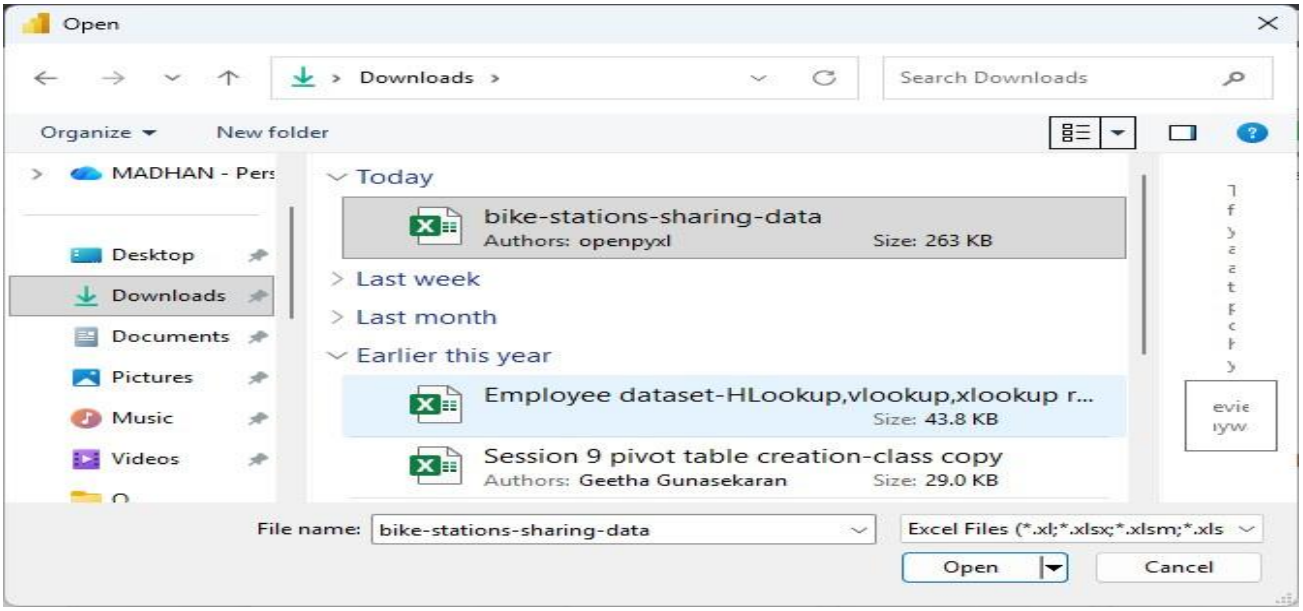
- **Data Cleaning & Transformation:** create separate id for Contractor make separate sheets using id.
- **Filtering & Sorting:** Organized data to focus on relevant records.

7. Data Modelling and DAX (Power BI)

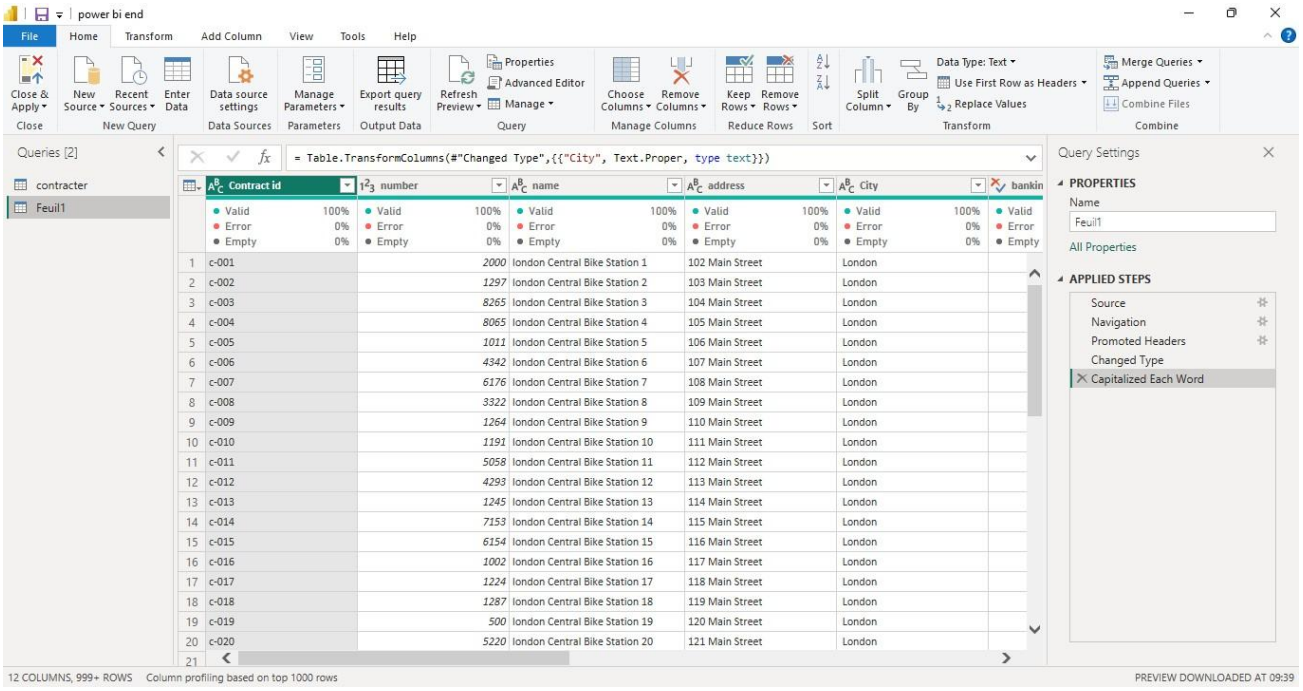
- Data Model: Established relationships between tables, defined cardinality, and created lookup tables where necessary.

SCREENSHOT OF DATA MODELLING AND IMPORT DATA





DATA CLEANING SCREENSHOT



DAX Measures:

Implemented DAX formulas for key metrics, such as total BIKES,Total stands and total KM.

FORMULA:

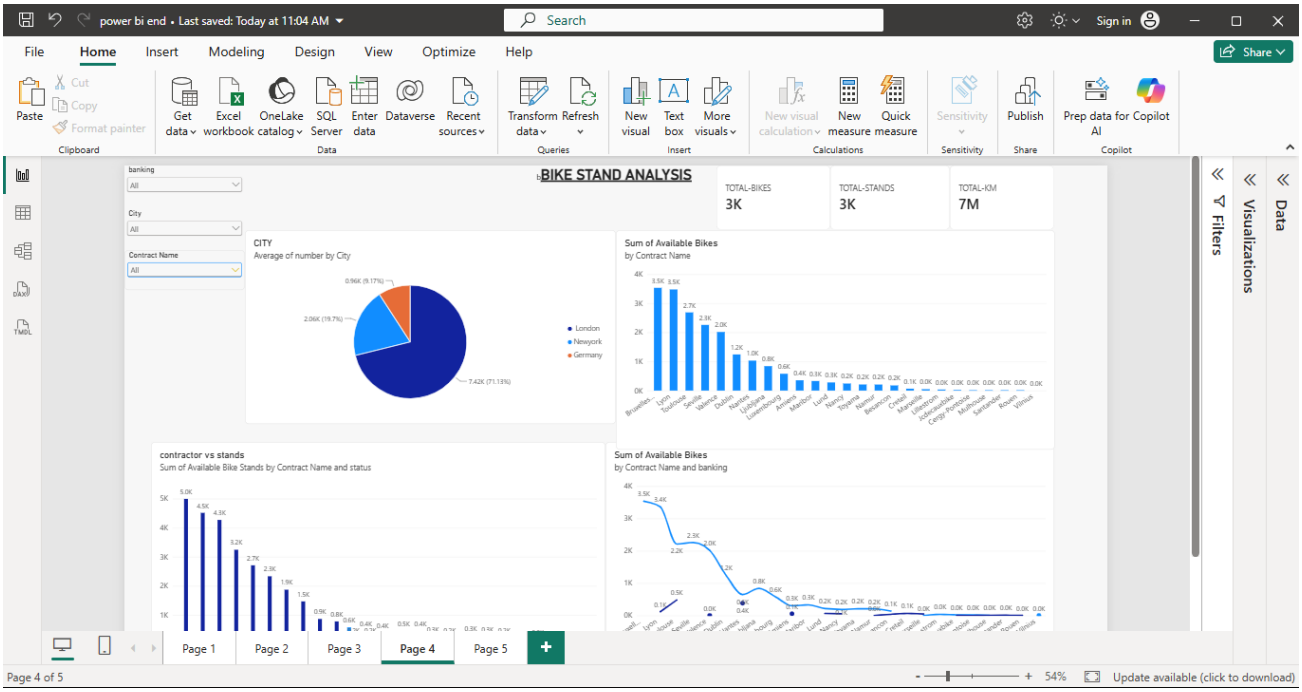
- Total BIKE formula: `TOTAL-BIKES = COUNT(Feuil1[Available Bikes])`
- Total-stands: `TOTAL-STANDS = COUNT(Feuil1[Available Bike Stands])`
- Total-KM: `TOTAL-KM = SUM(Feuil1[number])`

7. Analysis and Visualizations (Power BI):

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights.

- Make your Report interactive Layout using **slicers.**
- **Create a consolidated Report /Dashboard.**

SCREENSHOT OF DASHBOARD /REPORT



9. Insights & Conclusions

- **Key Findings:**

➢ The dashboard helps identify total available bikes and Total KM
➢ City wise analysis helps profit or loss.
➢ Payment status analysis helps understand successful and pending payments.
➢ Monthly analysis helps identify sales trends over time.

Provide the analysis insights:

1. Descriptive Analysis

- Analyzed total km, total bikes,total stands.

2. Diagnostic Analysis

- Compared sales across different contractor and bike stands.
- Analyzed payment status to understand payment-related differences.

3. Predictive Analysis

- Used historical monthly sales trends to understand possible future sales patterns.
- Identified potential changes in sales based on previous trends.

4. Prescriptive Analysis

- Focus on products and categories with lower sales to improve performance.
- Improve customer engagement to increase sales.

10. Conclusions:

The Product Sales Analysis project helps understand overall sales performance using Power BI. It provides useful insights contractor behaver , monthly sales trends, and payment status. Interactive dashboards and visualizations make it easier to identify sales patterns and business performance. This analysis helps businesses make better decisions, improve sales, and plan future strategies.